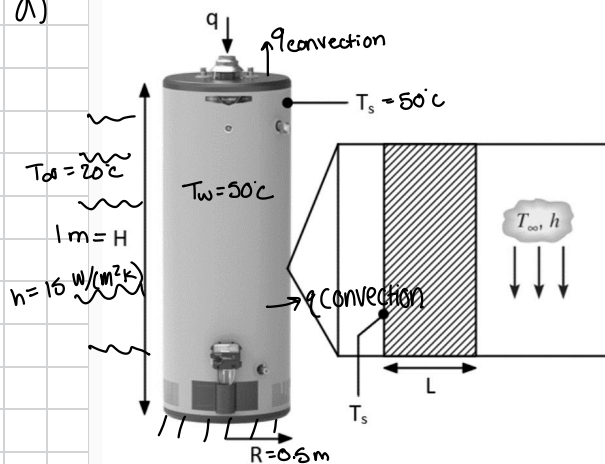


Known: Steady state, $R=0.5$, $H=1\text{m}$ $T=50^\circ\text{C}$ $T_s=50^\circ\text{C}$ $T_a=20^\circ\text{C}$ $h=15\text{ W}/(\text{m}^2\cdot\text{K})$
bottom insulated, radiation negligible

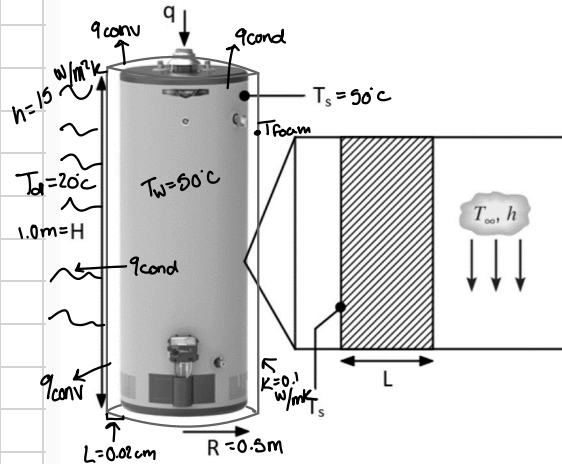
Find: a) q needed for 50°C
b) q needed for 50°C w/ changes
c) role of thermal insulation

Schematic

a)



b)



Assume: ① bottom insulated ② steady state ③ Heater wall thin w/ $\uparrow$ therm. conductivity ④ radiation negligible

Analysis:

$$\begin{aligned} \text{a) } E_{in} &= E_{out} \rightarrow E_{in} = q \quad E_{out} = hA(T_s - T_a) \\ A &= \pi R^2 + 2\pi RH = \pi(0.5\text{m})^2 + 2\pi(0.5\text{m})(1\text{m}) = 3.927\text{m}^2 \\ q &= (15\text{ W}/(\text{m}^2\cdot\text{K}))(3.927\text{m}^2)(50^\circ\text{C} - 20^\circ\text{C}) = (15)(3.927)(30\text{K}) = 1767.2\text{W} \end{aligned}$$

$$\begin{aligned} \text{b) } E_{in} &= E_{out} \rightarrow E_{in} = hA(T_s - T_a) = \frac{kA(T_s - T_f)}{L} = q \\ hA(T_s - T_a) &= \frac{kA(T_s - T_f)}{L} \\ LhT_s - LhT_a &= kT_s - kT_f \\ LhT_f + kT_f &= kT_s + LhT_a \\ T_f(Lh + k) &= kT_s + LhT_a \\ T_f &= \frac{kT_s + LhT_a}{Lh + k} = \frac{(0.1\text{ W}/(\text{m}\cdot\text{K}))(50 + 273.15\text{K}) + (0.02\text{m})(1\text{m})(20 + 273.2)}{(0.02\text{m})(1\text{m}) + (0.1\text{ W}/(\text{m}\cdot\text{K}))} = 300.65\text{K} \\ q &= \frac{hA(T_w - T_f)}{L} = \frac{(15\text{ W}/(\text{m}^2\cdot\text{K}))(3.927\text{m}^2)(50 + 273.15\text{K} - 300.65\text{K})}{0.02\text{m}} = 441.79\text{W} \end{aligned}$$

$$\begin{aligned} \text{c) } \text{w/o insulation} &\rightarrow q = 1767.2\text{W} \quad \text{w/insulation} = 441.79\text{W} \\ \frac{1767.2 - 441.79}{1767.2} \times 100 &= 75\% \end{aligned}$$

Using insulation saves 75% of the energy needed to operate the tower.